

A Project report on

HYDRAULIC LIFT

Submitted in partial fulfillment of the requirements for award of the
degree of

Bachelor of Technology in

Computer science and Engineering By:

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Under the Esteemed Guidance of **Mrs. K. Sujatha** **Head of
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**DADI INSTITUTE OF
ENGINEERING & TECHNOLOGY
(AN AUTONOMOUS INSTITUTE)**

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CERTIFICATE

This is to certify that the project report entitled **“Hydraulic Lift”** submitted By requirements for award of the Degree of **Bachelor of Technology** in **Computer Science Engineering**, from Dadi Institute of Engineering & Technology(A), Anakapalle affiliated to JNTUGV, accredited by NAAC with 'A' grade is a record of Bonafide work carried out by them under my guidance and supervision.

PROJECT

GUIDE

HEAD OF THE DEPARTMENT

EXTERNAL EXAMINER

ACKNOWLEDGEMENT

We would like to express our gratitude to our project guide, **K. Sujatha, Professor & Head of the Department of the Computer Science & Engineering**, for her guidance and assistance throughout the development of this project. Without her supervision, support, and encouragement, we would not have gained awareness of many new things during our project.

We convey our heartfelt thanks to **Dr. K. Sujatha, Professor & Head of the Department of Computer Science & Engineering**, for motivating us to

We would like express our deep sense of gratitude to **Honorable Chairman, Sri Dadi Ratnakar**, of Dadi Institute of Engineering & Technology, for providing necessary facilities to carry out our project work successfully.

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DECLARATION

We hereby declare that the project entitled **“HYDRALIC LIFT”** is submitted in partial fulfilment of the requirements for the award of Bachelor of Technology in **Computer Science & Engineering** under esteemed supervision of **Mrs.K. Sujatha, Professor & Head of the Department**. This is a record of work carried out by us and results embodied in this project report have not been submitted to any other university for the award of any Degree.

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ABSTRACT:

Hydraulic lift is the process by which some deep-rooted plants take in water from lower soil layers and exude that water into upper, drier soil layers. Hydraulic lift is beneficial to the plant transporting the water, and may be an important water source for neighbouring plants. Recent evidence shows that hydraulically lifted water can promote greater plant growth, and could have important implications for net primary productivity, as well as ecosystem nutrient cycling and water balance. The upward motion is achieved by the application of pressure to outside of the lowest set of support elongating the crossing pattern and propelling the work platform vertically. In our case our lift was needed to be designed a portable and also work without consuming any electric power so we decided to use a hydraulic hand pump to power the cylinder also such design can make the lift more compact and much suitable for medium scale work. This paper describes the complete study of components (hydraulic cylinder, scissor arms, spacing shaft and platform), election of materials and analysis the dimensions of components. Further fabrication of all the parts and assembly is carried out. Hydraulic lift is become very essential part of human life and also for companies that needs specific lifting elevators, that huge requirement is very customizable and also company oriented. Product design becomes complex because every products function is different from every aspect.

Keywords:- Hydraulic Scissor Lift, Pantograph, Hand pump, Vanishes stresses, Hydraulic power pack.

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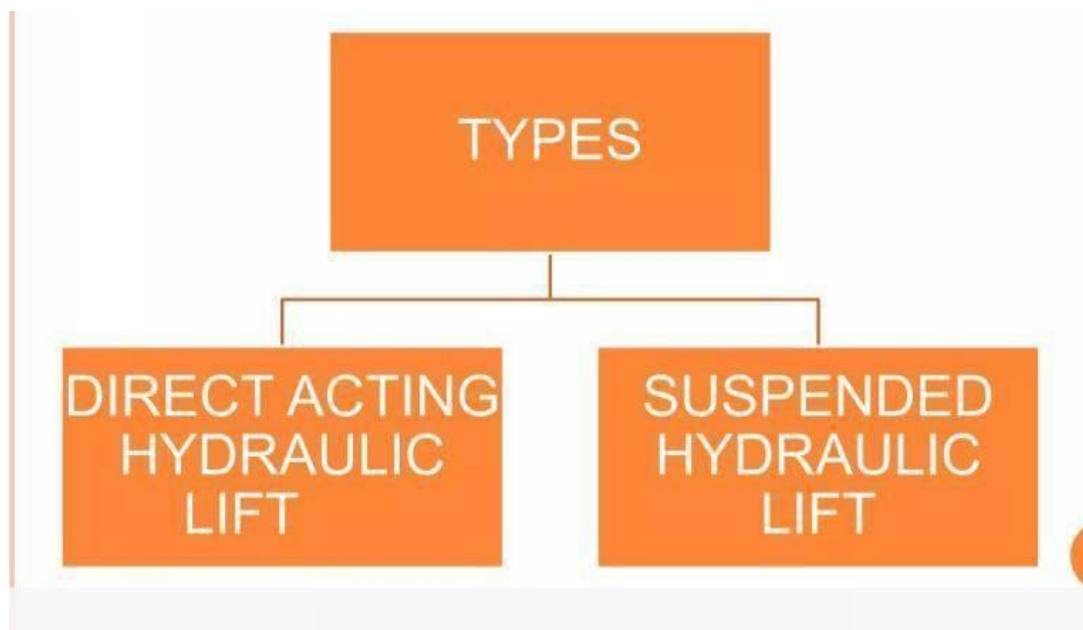
Aim:

HYDRAULIC LIFT

INTRODUCTION:

Hydraulic is a technology and applied science using engineering, chemistry, and other sciences involving the mechanical properties and use of liquids. At a very basic level, hydraulics is the liquid counterpart of pneumatics, which concerns gases. Fluid mechanics provides the theoretical foundation for hydraulics, which focuses on the applied engineering using the properties of fluids. In its fluid power applications, hydraulics is used for the generation, control, and transmission of power by the use of pressurized liquids. Hydraulic topics range through some parts of science and most of engineering modules, and cover concepts such as pipe flow, dam design, fluidics and fluid control circuitry. The principles of hydraulics are in use naturally in the human body within the vascular system and erectile tissue. Free surface hydraulics is the branch of hydraulics dealing with free surface flow, such as occurring in rivers, canals, lakes, estuaries and seas.

Hydraulic Lifts: Types



Direct Acting Hydraulic Lift:

- When fluid under pressure is forced into the cylinder, the ram gets a push upward. The platform carries loads or passengers and moves between the guides. At required height, it can be made to stay in level with each floor so that the goods or passengers can be transferred.
- In direct acting hydraulic lift, stroke of the ram is equal to the lift of the cage.

Suspended Hydraulic Lift:

Advantages and Disadvantages of Hydraulic Lifts

ADVANTAGES:

1. Hydraulic lifts are cheaper to install than other elevator types.
2. They occupy less space in a building, requiring almost 10% less area for the lift shaft.
3. They are highly effective with heavy loads, as the hydraulic power provides a far greater lifting strength.
4. There is no need for an overhead machine room. Similarly, overhead structural requirements aren't needed either. The load of the elevator is distributed to load-bearing walls.
5. You have the option of using a remote machine room.
6. Despite the popularity of the hydraulic elevator system, it does come with its own set of drawbacks.

Disadvantages:

1. While the disadvantages really are few and far between, just some of them include:
2. The relatively slow speed. A hydraulic lift cannot move faster than 150 feet per minute.
3. Oil in the piston system can overheat fairly quickly, which is why an efficient machine-room control is important.
4. Hydraulic oil can leak out of the pumping system and into the ground at the base of the elevator. This can lead to the pollution of groundwater.

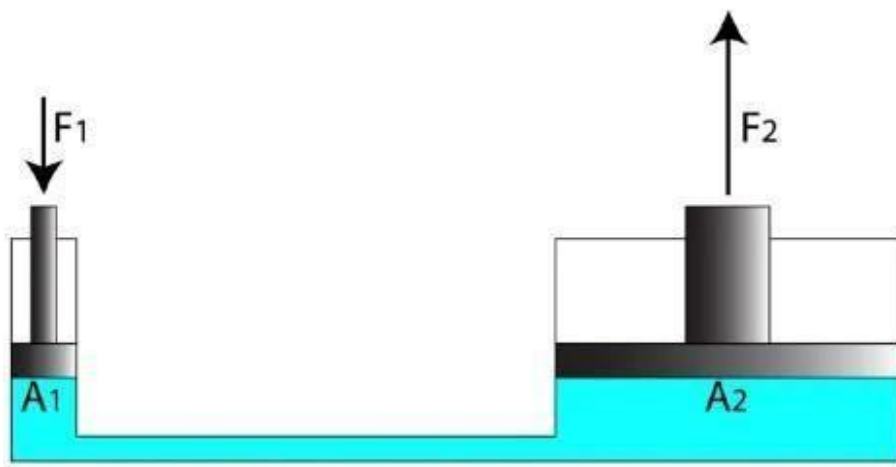
APPARATUS REQUIRED

1. Cardboard
2. 10ml Syringes
3. Rubber tubes
4. A toy car
5. Glue gun

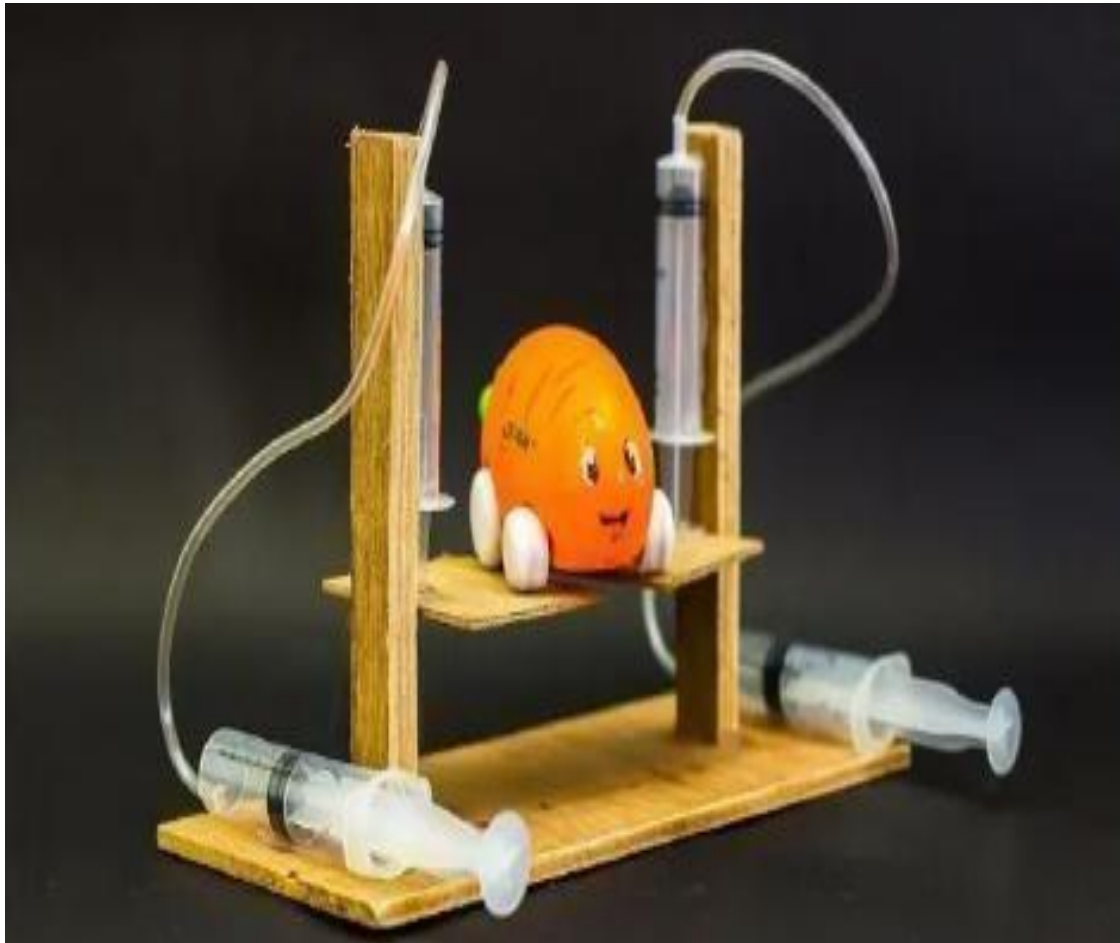


Working principles of hydraulic lift:

Hydraulic lift works on the principle of Pascal's law which states that if any force is applied to any point of a confined fluid, then the pressure is equally and undiminishable transmitted through out the liquid.



Layout Diagram:



Procedure:

- 1) Take a cardboard base and glue two cardboard bar in it and stick the syringe
- 2) Fill a beaker with water and add some amount of a colour if wanted, to make it attractive when in use.
- 3) Take four 10ml syringe and fill 2 of the syringes fully with the colour water made in the beaker, fix rubber pipes to the end of the syringe and push the plunger of the syringe so that the water will pass through the tip of the syringe and push the plunger till the colour water reach to the rubber tube.
- 4) Take the other two syringes fill it up to its half volume with colour water from the beaker. So that when connected with other end of the rubber tube, and when pushed the plunger of one of the syringes, then the plunger of other syringe will be pulled out due to accumulation of liquid in another syringe.

5) Stick two 10ml syringe at the side of each cardboard

bars using glue gun and stick one cardboard sheet each with each plunger of the syringe and stick other two syringes to the main cardboard base so that when pressure is applied to the syringe cardboard base will move up and down.

6) Place the toy car at the base and apply the pressure at the syringes, as the result the toy car is lifted up and down.



OBSERVATION:

As the water is pushed or pulled through the syringe the cardboard base on which the toy car is present will go down or will be lifted up.

PRECAUTIONS:

1. The weak cardboard should not be used.
2. The load carrying base should not be weak it has to be strong enough to carry the load.
3. Properly stick the cardboard using glue gun and also use the glue gun safely.
4. Syringes should not be filled with liquid in excess amount.

APPLICATIONS:

1. Wheel chair lift and also in Industrial hydraulic applications
2. Material handling.
3. Truck trailer.
4. Used in heavy vehicles like-JCB.

CONCLUSION:

- Hydraulic lift is a device which is used to carry large loads with the application of small force.
- A modern way to replace levers and pulleys.
- Ability to control the speed of the base in one direction only.